

Module 3: Technology and tools for Integrated Engineering projects



Lecture NO: 21

Introduction to Automation and Smart Technologies



AUTOMATION

Definition: It is technology concerned with the application of Mechanical, electronic & computer-based systems to operate and control production in order to improves productions.



Robotics in Construction: Automation



automation is changing the mechanical engineering industry



TYPE OF AUTOMATION

Automated production systems are classified into three basic types:

- 1) Fixed Automation
- 2) Programmable Automation
- 3) Flexible Automation

Source: <https://nptel.ac.in/courses/112102011>



1) Fixed Automation

Fixed automation is a type of automation where the process of manufacturing stays fixed by the way it is configured, following a fixed sequence of automated processes. An example of this is flow production, where products are continuously being made. This is often also known as “hard automation

Advantages of fixed automation include:

- High levels of production
- Consistent quality in production
- Low cost per unit produced

Disadvantages of fixed automation include:

- High initial cost
- Difficult to accommodate changes



2) Programmable Automation

Programmable automation allows the production equipment and automation to be altered to changing needs. This is done by controlling the automation through a program, which can be coded in certain ways for the automation to change the sequence of automation.

It's used more commonly in low to medium levels of production, often being most suitable for batch production

Advantages include:

- Flexibility to change products if needed
- Suitable if batch production is required

Disadvantages include:

- Expensive for equipment
- Lower production levels
- Often time-consuming to change product

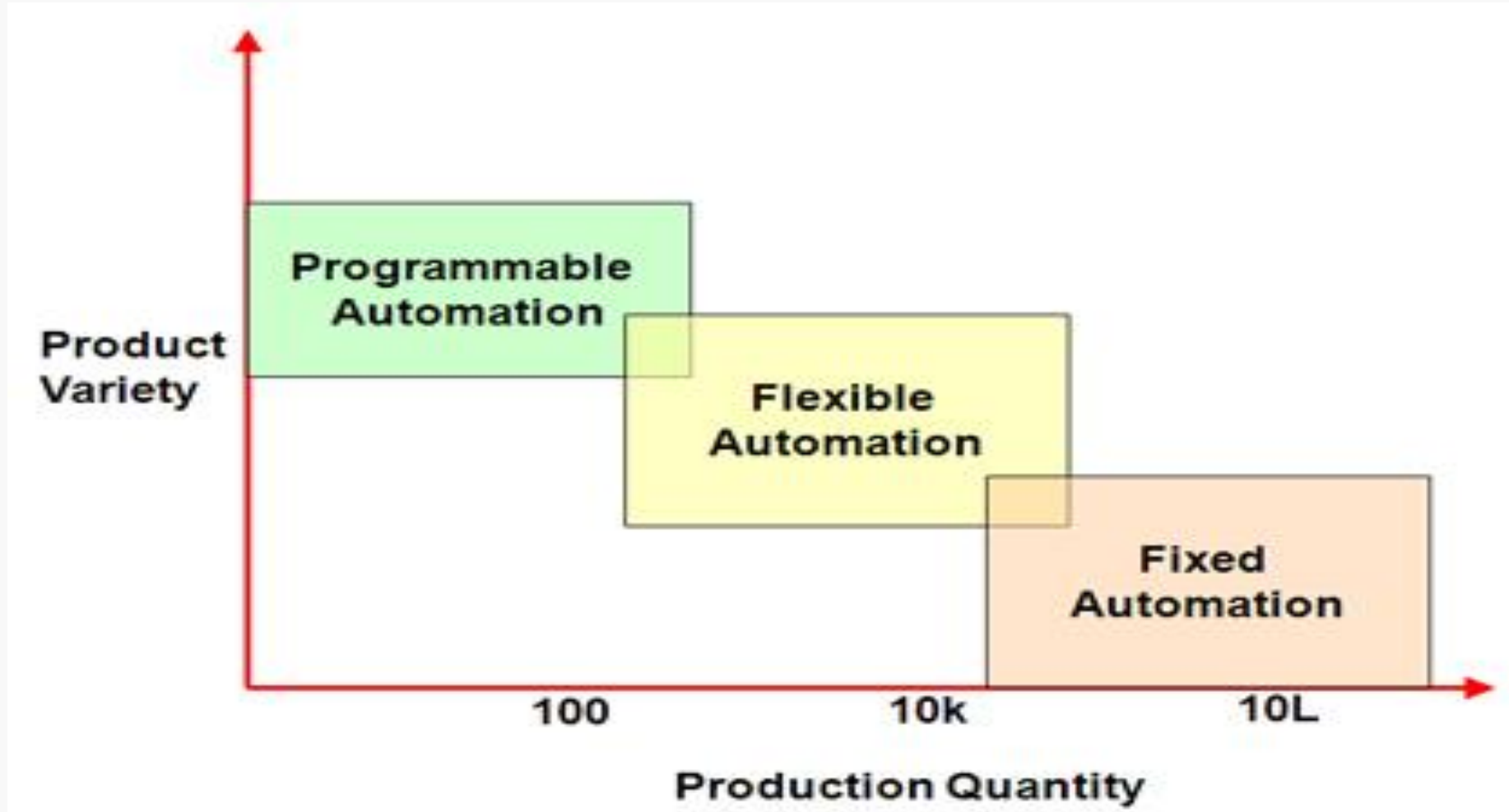


3) Flexible Automation

- Extension of programmable automation.
- Produce variety of parts with virtually no time lost for changeovers from one part style to next.
- No production time lost in set up and reconfiguration.
- System can produce various mixes of parts.
- High Initial cost and low production rate
- Example : flexible manufacturing system (fms) in automation



Three Type of Automation



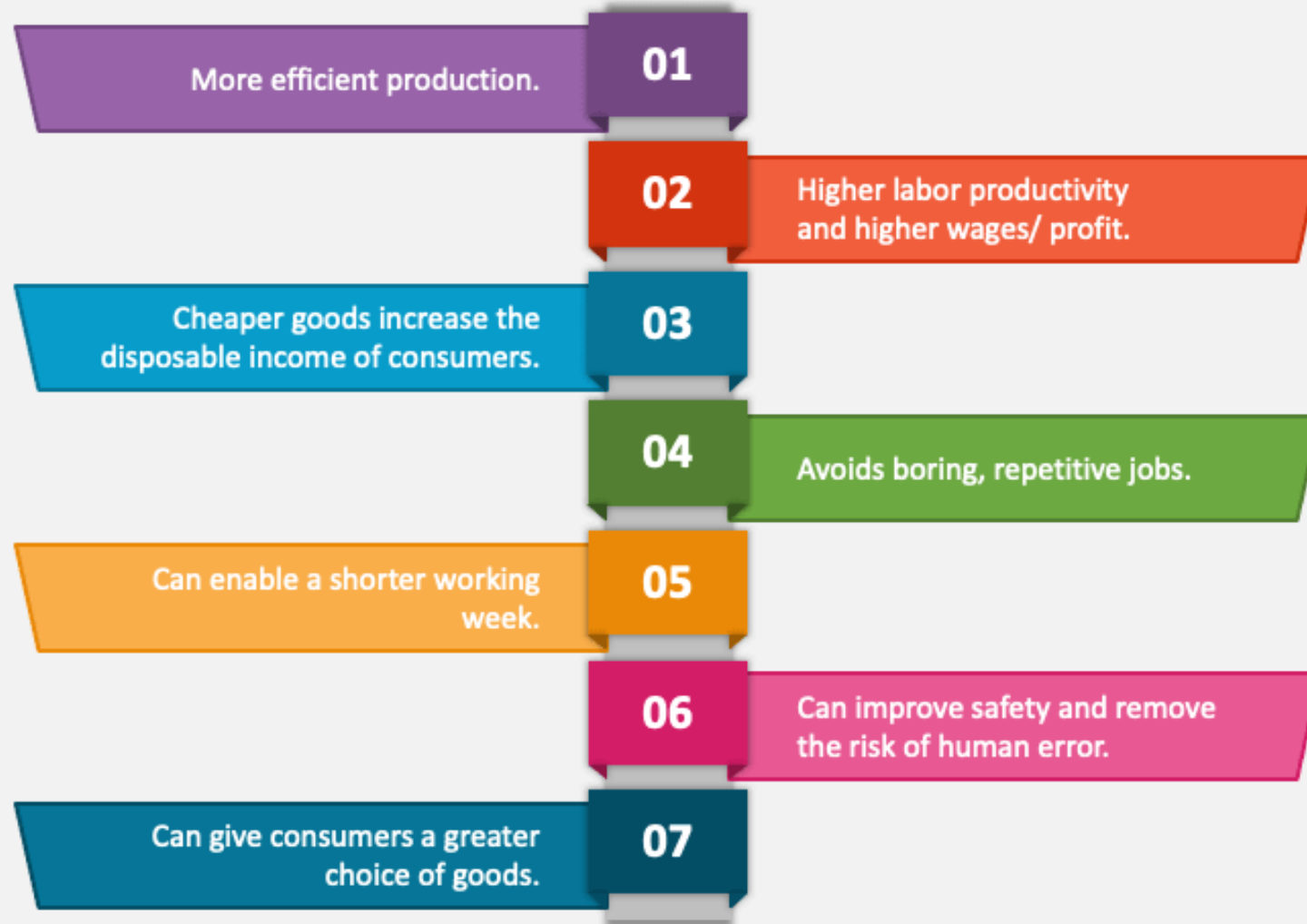
AUTOMATION

It Includes:

- 1) Automated machine tools.
- 2) Automated assembly machines
- 3) Industrial robots.
- 4) Automated material handling & storages system
- 5) Automated inspection system for quality control
- 6) Feedback control & computer process control
- 7) Computer integrated system for planning, data collection decision making.



BENEFITS OF AUTOMATION



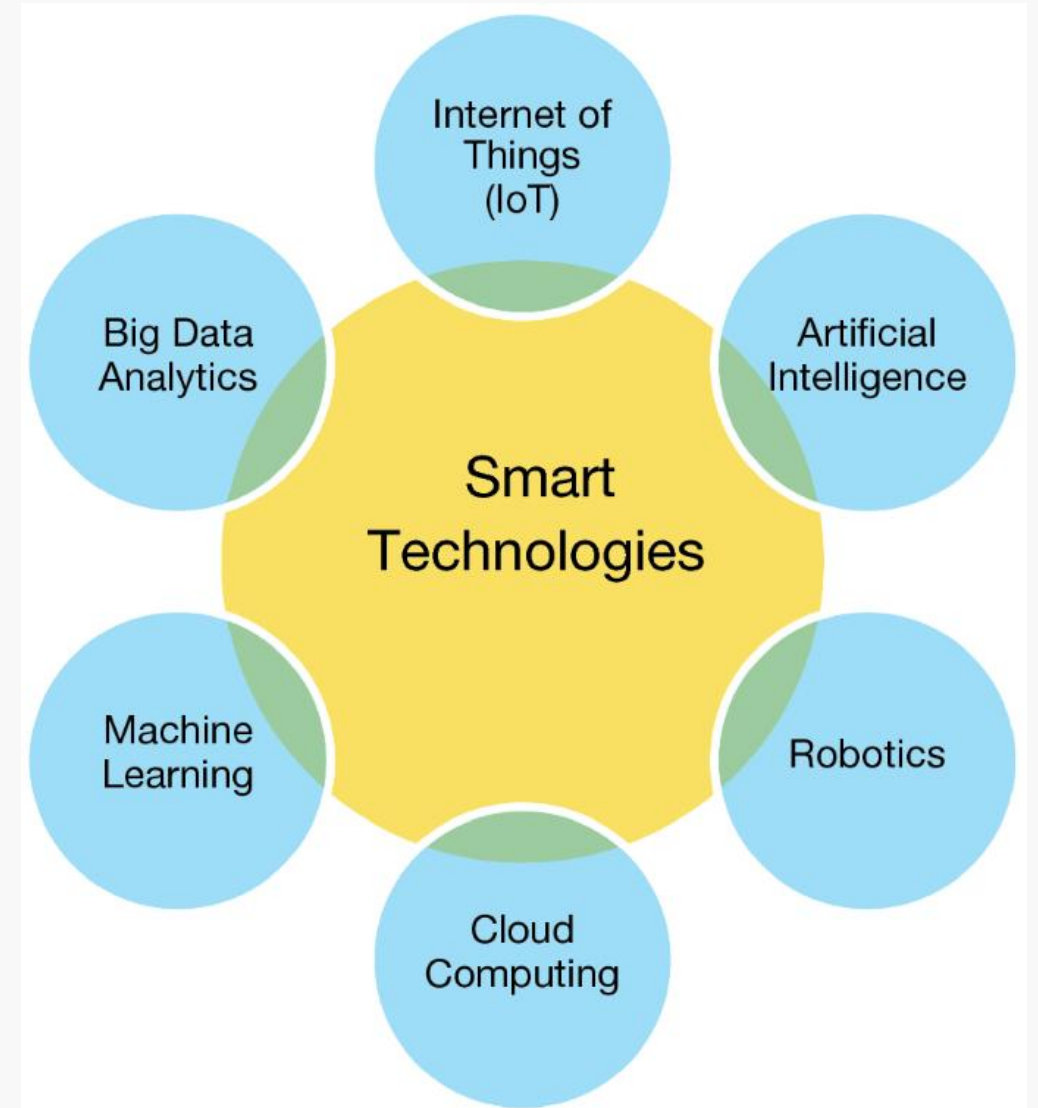
What Are Smart Technologies?

- Technologies that sense, process, and respond intelligently
- Combine AI, IoT, big data, and automation
- Improve efficiency, performance, and user interaction
- Self-learning and adaptive systems

Key smart technologies

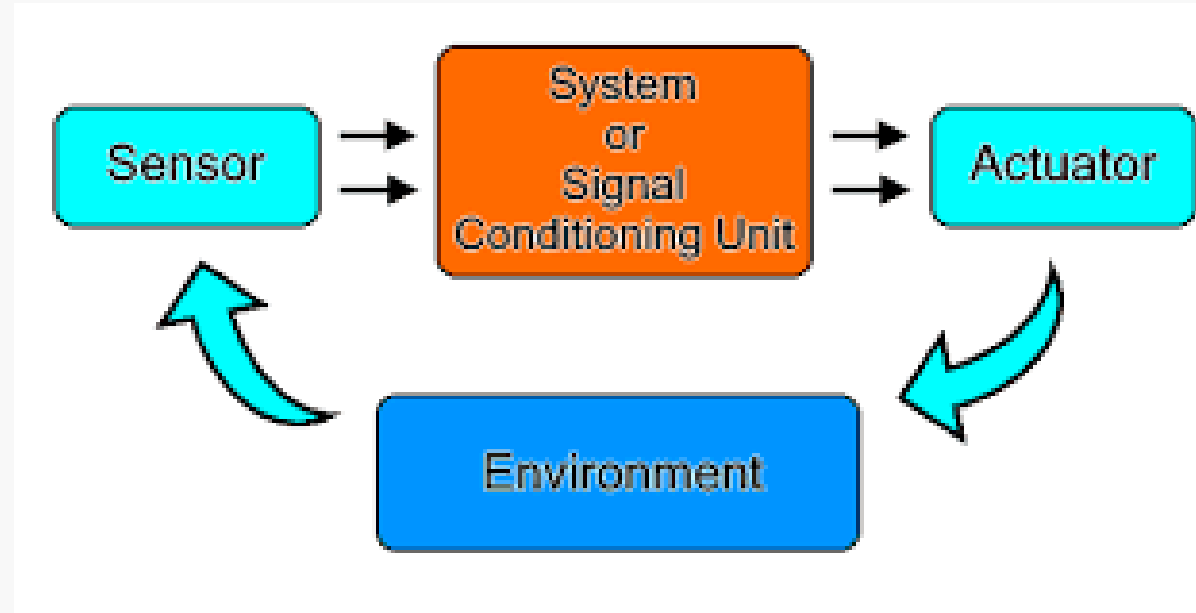
Internet of Things (IoT) – Devices connected to the internet

- Artificial Intelligence (AI)** – Smart decision-making
- Machine Learning (ML)** – Adaptive behavior through data
- Cloud & Edge Computing** – Fast, scalable processing
- Blockchain** – Secure data sharing



Core Components of smart technologies

- **Sensors & Actuators** – Gather real-world data
- **Connectivity** – Wi-Fi, Bluetooth, 5G, cloud
- **Data Processing** – Edge computing, AI/ML
- **User Interfaces** – Apps, dashboards, voice, etc.



sensor: Sensor is a device that detects or measures a physical property (like temperature, light, motion, etc.) and converts it into a readable electrical signal

actuator : An **actuator** does the reverse. It **converts electrical signals into physical action**. It receives commands from the system (like turning on a motor or opening a valve)



Applications of Smart Technologies

- **Smart Homes** – Automation, security, energy saving
- **Smart Cities** – Traffic management, waste control, surveillance
- **Healthcare** – Remote monitoring, diagnostics
- **Industry (4.0)** – Robotics, predictive maintenance
- **Agriculture** – Precision farming, smart irrigation



Questions for practice

1. Define automation -----2M
2. List the types of automation -----2M
3. List any four benefits of automation----- 2M
4. explain the following type of automation with suitable examples -----10 M
a) Fixed Automation b)Flexible Automation
5. Compare the three type automation with the help of graph, Production variety Vs production quantity---5M
6. Define smart technologies-----2M
7. List any four types of smart technologies-----2M
8. List any four applications of smart technologies-----2M
9. With the help of neat sketch explain the Core Components of smart technologies-----10M



Lecture No: 22

Fundamentals and Characteristics of IoT

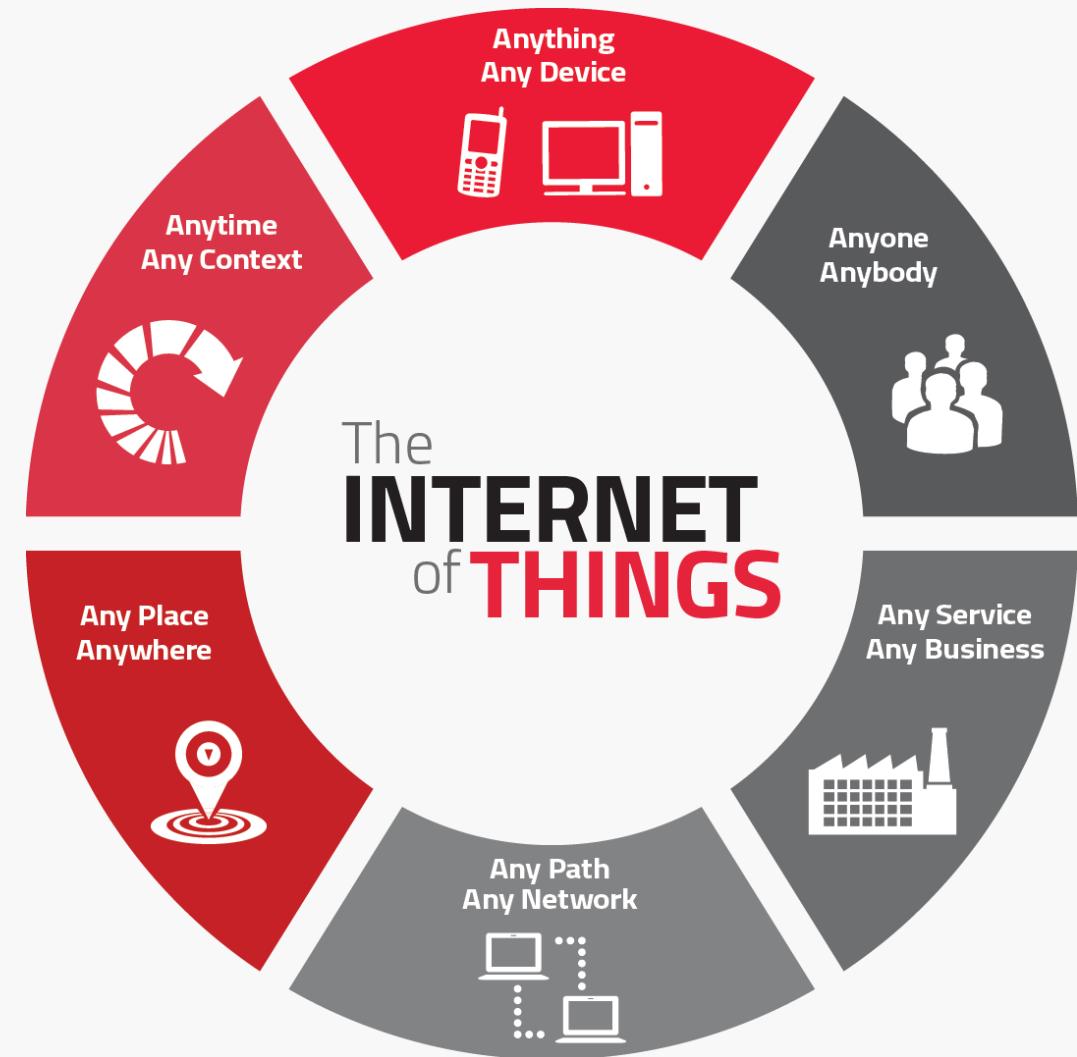


Definition of IoT:

The Internet of Things (IoT) refers to a vast network of physical objects, or "things," that are embedded with **sensors, software, and other technologies** to **connect and exchange data over the internet**. These interconnected devices, ranging from everyday household items to industrial machinery, can collect and share data, enabling them to interact with each other and with users.

Fundamentals of IoT

IoT systems allow users to **achieve deeper automation, analysis, and integration** within a system. They improve the reach of these areas and their accuracy. IoT utilizes existing and **emerging technology** for sensing, networking, and robotics.



➤ **Sensors and Devices:**

These are the physical objects that collect data from the environment or perform actions. Examples include temperature sensors, motion detectors, cameras, and actuators.

➤ **Connectivity:**

This refers to the network infrastructure that allows devices to communicate with each other and with the cloud. Technologies like Wi-Fi, cellular networks, Bluetooth, and Low range wide area network are used for connectivity.

➤ **Data Processing:**

Collected data needs to be processed and analyzed to extract meaningful insights. This involves cloud computing, edge computing, and data analytics techniques.

➤ **User Interface:**

A user interface (UI) allows users to interact with the IoT system, view data, and control devices. This can be a mobile app, a web dashboard, or other interface.



Sensors

Collecting data



Connectivity

Sending data to cloud



Data Processing

Making data useful



User Interface

Delivering information to user



Characteristics of IoT:

The fundamental characteristics of the IoT are as follows :

1. **Connectivity** –IoT devices need to be connected to a network (internet, LAN, or other communication protocols) to exchange data.
2. **Sensing** –IoT devices use sensors to collect real-time data from the physical environment (e.g., temperature, motion, pressure)
3. **Data Processing** –Collected data is processed locally (edge computing) or remotely (cloud computing) to derive meaningful insights.
4. **Automation & Control** –IoT systems can automate actions without human intervention, based on pre-set rules or AI decision-making.
5. **Scalability** –IoT networks can scale up to include millions of devices while maintaining performance and reliability.



Characteristics of IoT:

- 6. Interoperability** –Devices from different manufacturers should work together using common communication standards and protocols.
- 7. Security** –Protecting data, devices, and networks from cyber threats is essential in IoT due to the large attack surface.
- 8. Real-Time Operation** –IoT often requires instant or near-instant responses for applications like healthcare monitoring or autonomous vehicles.
- 9. Energy Efficiency** –Many IoT devices are battery-powered, so low-power consumption is crucial for long-term operation.
- 10. Intelligence** –By integrating AI and machine learning, IoT systems can predict trends, optimize operations, and improve decision-making.



applications of IoT

Transport & Logistics  Fleet management, Goods tracking	Utilities  Smart metering, Smart grid management	Smart cities  Parking sensors, Waste management, etc.	Smart building  Smoke detector, Home automation
Consumers  Wearables Kids/senior tracker	Industrial  Process monitoring & control, Maintenance monitoring	Environment  Food monitoring/alerts, Environmental monitoring	Agriculture  Climate/agriculture monitoring, Livestock tracking



**Design and drafting using digital modeling for
components, systems, and structures**

Lecture No: 23
Auto CAD and BIM: Advantages and applications



Introduction

- Design and drafting tools are essential for **creating, analyzing, and managing** engineering and architectural models
- Enable **precision, efficiency, and collaboration** in professional projects
- Widely used across **civil, mechanical, electrical, and architectural domains**



AutoCAD

- AutoCAD is a **computer-aided design (CAD)** software developed by **Autodesk**, widely used by engineers, architects, and designers to create precise **2D drawings** and **3D models**.

Key Features:

- **2D Drafting & Drawing Tools** – for detailed plans and layouts.
- **3D Modeling & Visualization** – surfaces, solids, and meshes.
- **Layer Management** – to organize different elements.
- **Dimensioning Tools** – accurate measurements and annotations.



Applications of AutoCAD:

- **Civil Engineering** – site plans, road layouts, drainage systems.
- **Mechanical Engineering** – parts design, assemblies, manufacturing drawings.
- **Architecture** – floor plans, elevations, building sections.
- **Electrical & Plumbing** – schematics and wiring layouts.

Advantages of AutoCAD :

- High precision and accuracy
- Saves time over manual drafting
- Easy editing and modification
- Widely accepted industry standard



BUILDING INFORMATION MODELLING

- Building Information Modelling (BIM) is a very broad term that describes the process for specifying, creating, and managing digital information about a built asset such as a building, bridge, highway or tunnel.
- BIM integrates structured, multi-disciplinary data to produce a digital representation of an asset across its lifecycle, from planning and design to construction and operations.
- The process of BIM supports the creation of intelligent data that can be used throughout the lifecycle of a building or infrastructure project.



BUILDING INFORMATION MODELLING - BENEFITS



Plan

Inform project planning by combining reality capture and real-world data to generate context models of the existing built and natural environment.



Design

During this phase, conceptual design, analysis, detailing and documentation are performed. The preconstruction process begins using BIM data to inform scheduling and logistics.



Build

During this phase, fabrication begins using BIM specifications. Project construction logistics are shared with trades and contractors to ensure optimum timing and efficiency.



Operate

BIM data carries over to operations and maintenance of finished assets. BIM data can be used down the road for cost-effective renovation or efficient deconstruction too.

1. Visualize Projects in Preconstruction
2. Better Coordination and Clash Detection
3. Cost and resource savings due to Model-Based Cost Estimation (5D BIM)
4. Greater efficiency and shorter project lifecycles due to Improved Scheduling/Sequencing
5. Improved communications and coordination
6. More opportunities for prefabrication and modular construction
7. Higher quality results

